

Theory Assignment - 01

Answer 1

Whenever a user opens a website, the browser communicates with the website's server. The browser sends an HTTP request, and the server receives it, processes the required information, and returns an HTTP response.

Working process:

1. The user types a website URL in the browser.
2. The browser sends an HTTP request to the server.
3. The server receives the request and processes it.
4. After processing, the server sends back HTML, CSS, data, or other required content.
5. The browser interprets the response and displays the webpage to the user.

Answer 2

1. HTML: It is responsible for creating the basic structure and content of a webpage, such as headings, text, images, and links.
2. CSS: It is used to control the appearance of a webpage, including colours, fonts, spacing, and layout.
3. JavaScript: It adds behaviour and interactivity to webpages, for example responding to a user's click.
4. Bootstrap: It is a front-end framework that provides pre-built styles and components for developing responsive websites.
5. jQuery: It is a JavaScript library that simplifies several common JavaScript tasks.
6. React.js: It is a JavaScript library mainly used to build user interfaces and dynamic web applications.

Answer 3

Client-side technology operates in the user's web browser. Common examples are HTML, CSS, and JavaScript.

Server-side technology operates on the web server and is responsible for processing data and handling server-related operations. Examples include PHP and Node.js.

In simple terms, client-side technology mainly deals with what the user sees and interacts with, whereas server-side technology performs the background processing.

Answer 4

Block-level elements generally begin on a new line and normally occupy the available horizontal width. They are commonly used to divide a webpage into separate sections.

Examples:

```
<div>  
<p>  
<h1>
```

These elements may also contain other HTML elements within them.

Answer 5

Semantic elements describe the meaning or purpose of the content they contain. Because of this, they make the structure of an HTML document clearer and easier to understand.

Some examples are <header>, <nav>, <main>, <section>, and <footer>.

They also improve the organisation of a webpage and help search engines better understand the page content.

Answer 6

```
<table border="1">
<tr>
  <th>Sr. No.</th>
  <th>Name</th>
  <th>Designation</th>
  <th>Department</th>
</tr>
<tr>
  <td>1</td>
  <td>John</td>
  <td>Manager</td>
  <td>Sales</td>
</tr>
<tr>
  <td>2</td>
  <td>Smith</td>
  <td>Sr. Manager</td>
  <td>Marketing</td>
</tr>
<tr>
  <td>3</td>
  <td>Jane</td>
  <td>Manager</td>
  <td>HR</td>
</tr>
</table>
```

Answer 7

```
<ul>
<li>River
  <ul>
    <li>Ganga</li>
    <li>Yamuna</li>
    <li>Brahmaputra</li>
    <li>Narmada</li>
  </ul>
</li>
</ul>
```

Answer 8

CSS can be applied in three different ways:

1. Inline CSS: The CSS is written directly inside an HTML element by using the style attribute.

Example: <p style="color:red;">Hello</p>

2. Internal CSS: The CSS rules are placed inside a <style> tag within the same HTML document.
3. External CSS: The CSS is written in a separate .css file, which is then linked to the HTML page. This approach is generally useful for larger websites.

Answer 9

HTML provides the basic structure of a webpage. CSS is used to control its visual presentation, such as colours, fonts, and layout. JavaScript is responsible for adding actions and interactivity. Therefore, HTML can be viewed as the structure, CSS as the presentation, and JavaScript as the behaviour of a webpage.

Answer 10

- a. Bootstrap: It provides ready-to-use CSS classes and interface components, helping developers create responsive websites more quickly.
- b. jQuery: It is a JavaScript library that became popular because it simplified tasks such as DOM manipulation, event handling, and other JavaScript operations.
- c. React.js: It is a JavaScript library used for building user interfaces. It supports the development of dynamic pages by organising the interface into reusable components.